# PARISHRAM 2.0

## CLASS 12TH 2026

## Electric Charges and Fields

PHYSICS LECTURE-8

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# Topics to be covered

1. Gauss's Law
2. Electric Field due to a Point charge (By Gauss's Law)
3. Electric Field due to an infinite linear charge distribution

$\frac{Kq}{r^2}$ !

$\Phi_E = \frac{q_{enc}}{\varepsilon_0}$

$\Phi_E = \vec{E} \cdot \vec{A}$

# Recap

1. Electric Flux
2.
3.
4.

$$\phi_E = \vec{E} \cdot \vec{A} = \frac{Q_{enclosed}}{\varepsilon_0}$$

# QUESTION

A sphere of **radius R** and **charge Q** is placed inside an imaginary **hollow** sphere of **radius 2R** whose centre coincides with the given **sphere.** The flux related to imaginary **sphere** is:

A $\frac{Q}{\epsilon_0}$

B $\frac{Q}{2\epsilon_0}$

C $\frac{4Q}{\epsilon_0}$

D $\frac{2Q}{\epsilon_0}$

$\Phi_E = \frac{Q_{enc}}{\epsilon_0}$

$\Phi_E = \frac{Q}{\epsilon_0}$



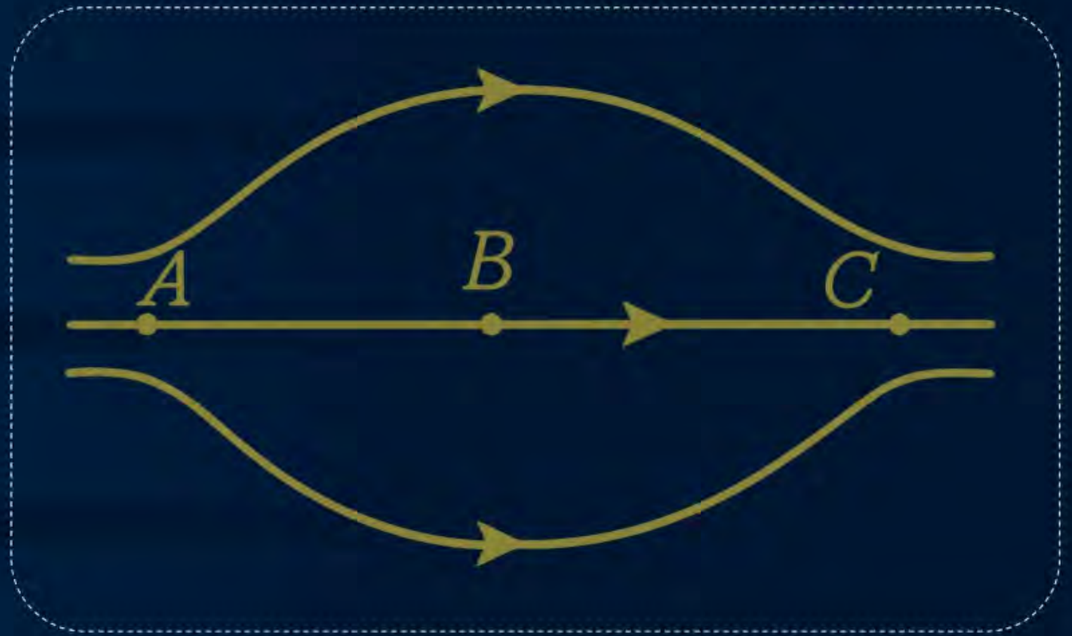

Imaginary hollow sphere



# QUESTION

The figure shows some of the **electric field** lines **corresponding** to an **electric field**. The figure suggests:

$E_A = E_C > E_B$




**A** $E_A > E_B > E_C$

**B** $E_A = E_B = E_C$

**C** $E_A = E_C > E_B$

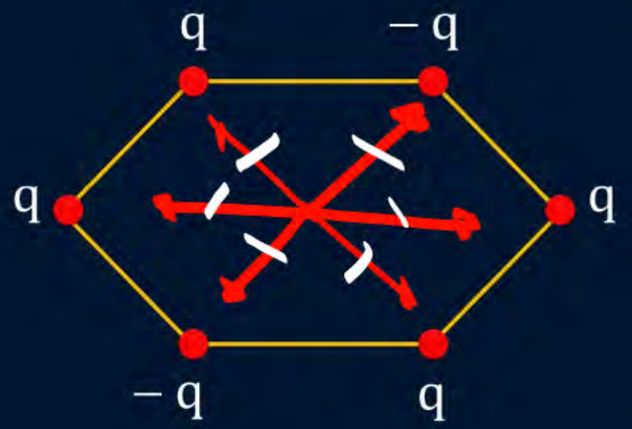
**D** $E_A = E_C < E_B$

# QUESTION

CHOCOLATE QUES.

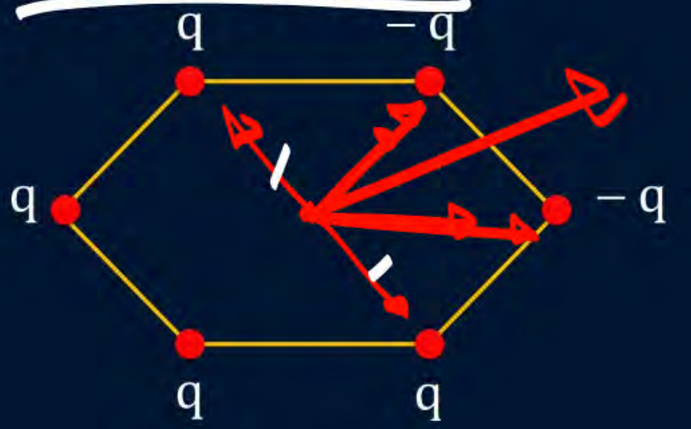
In which of the following cases the **electric field** at the centre is **not zero** ?

**A**

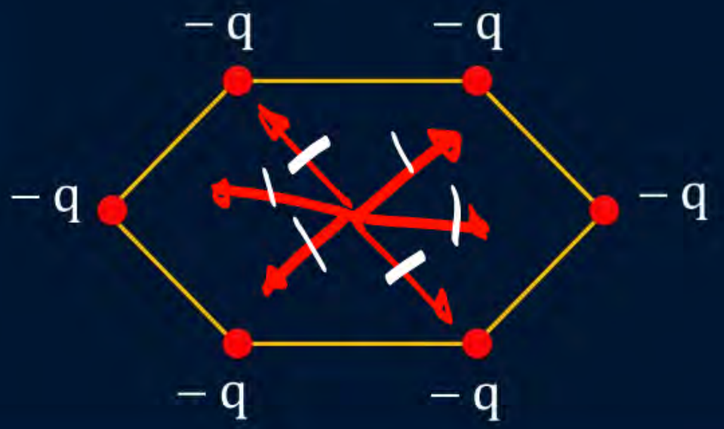



**B**

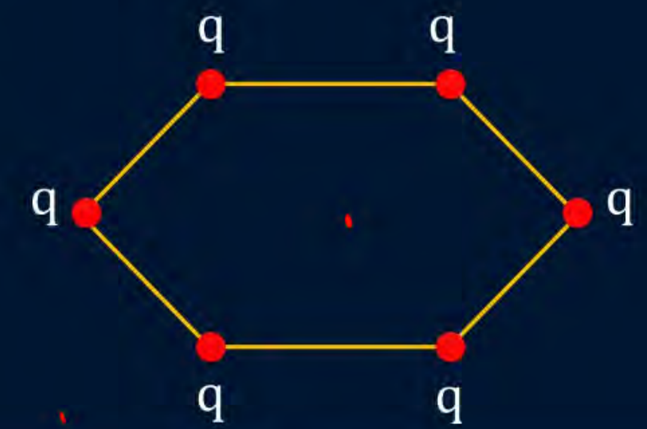
Non-zero




**C**




**D**





How I look up to that one student who asks “Class kab takk chalegi” in the initial two minutes of the class:

# QUESTION

IIT

Find the **flux** due to the **charge Q** through **Face-1** and **Face-2**.

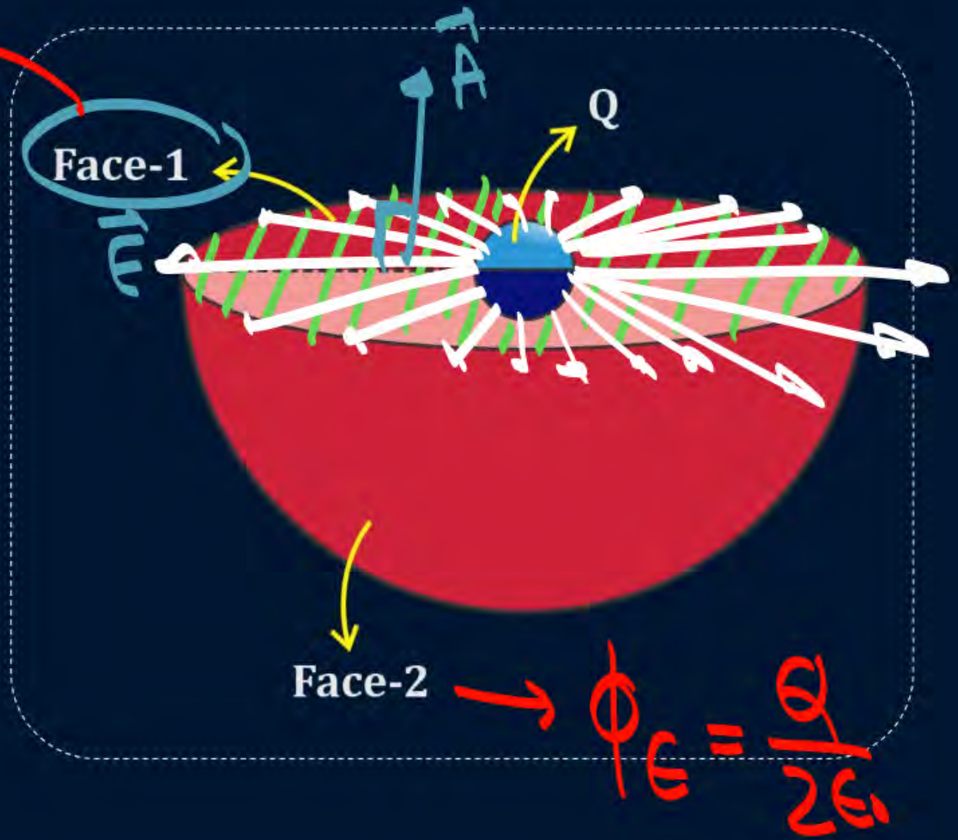
$Q_{enclosed} = \frac{Q}{2}$

**Face-2** $\boxed{(\phi_E)_2 = \frac{Q}{2\epsilon_0}}$

**Face-1** $(\phi_E)_1 = E \cdot A \cdot \cos(90^\circ)$

$= \boxed{0}$

$\phi_E = 0$




Face-2 $\rightarrow \phi_E = \frac{Q}{2\epsilon_0}$

# Gauss's Law or Gauss's Theorem

According to **Gauss's Law**, "the **net electric flux** through any closed **surface (S)** is equal to $\frac{1}{\varepsilon_0}$ times the **net charge (q)** enclosed by it. Mathematically, it can be written as:

$$\phi_E = \oint \vec{E} \cdot \vec{dS} = \frac{q}{\varepsilon_0}$$




Gauss's Law :-

$$\phi_E = \oint \vec{E} \cdot \vec{dS} = \frac{Q_{enclosed}}{\varepsilon_0}$$

# Derivation of Gauss's Law

(Derivation is incomplete without this diagram.)

Gauss's law :- $\Phi_E = \underbrace{\oint \vec{E} \cdot \vec{ds}}_{\text{LHS}} = \underbrace{\frac{Q_{enclosed}}{\epsilon_0}}_{\text{RHS}}$

**LHS** $\oint \vec{E} \cdot \vec{ds}$

$= \oint \left(\frac{kQ}{r^2}\right)(ds)\cos 0°$

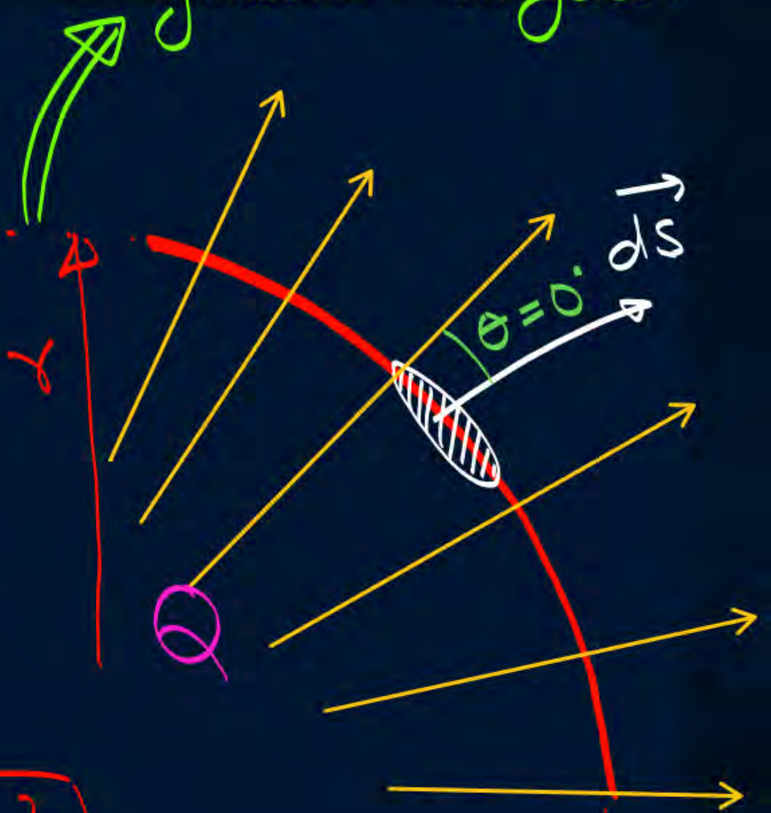
$= \left(\frac{1}{4\pi\epsilon_0}\right)\frac{Q}{r^2}\oint ds$ $\quad \oint ds \rightarrow S$

$= \frac{Q}{4\pi\epsilon_0 r^2}(4\pi r^2) = \boxed{\frac{Q}{\epsilon_0}} = \text{RHS}$

Total Surface area $= S$

$= \boxed{4\pi r^2}$

imaginary gaussian surface.



# Gaussian Surface :-




Que: Find the electric flux because of the charge Q at a distance r from it using Gaussian law?

Ans. $\phi_E = \frac{Q}{\epsilon_0}$

↳ Imaginary Gaussian Surface

# Characteristics of Gauss's Law




Characteristics of **Gaussian Surface** & Important Points Regarding **Gauss' Law** :

(i) Flux through **Gaussian** surface is independent of its **shape**.

(ii) Flux through **Gaussian** surface depends only on charges present **inside** it.

(iii) Flux through **Gaussian** surface is independent of position of **charges inside** it.

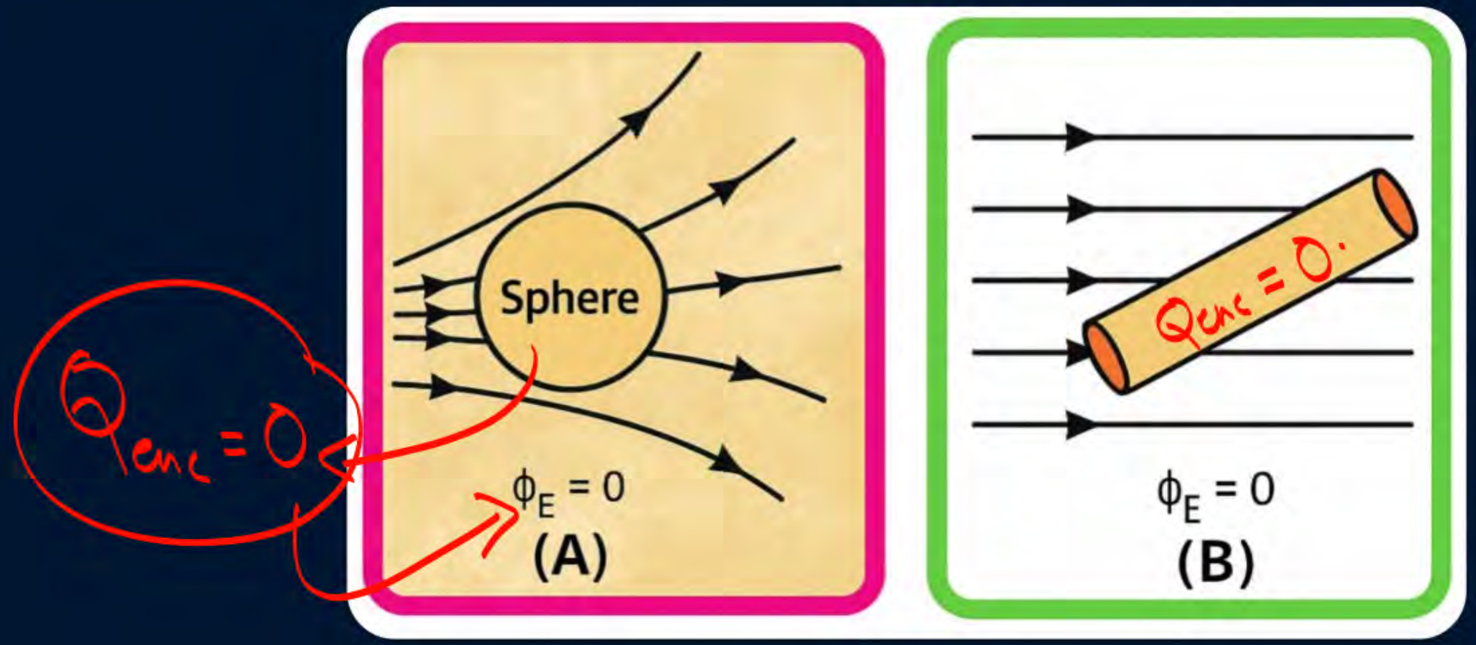
(iv) All of the charges present contribute to the intensity of the **electric** field at the **Gaussian** surface.

(v) Incoming flux is considered **negative** and **exiting flux** is considered **positive** in a closed **surface**.

(vi) In a Gaussian surface **$\phi = 0$ does not imply $E = 0$ but $E = 0$ implies $\phi = 0$.**

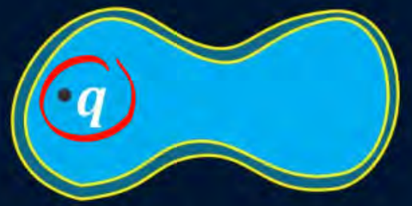
# Characteristics of Gauss's Law

(vii) (a) The total **flux** associated with a **closed body** that is not enclosing any charge will be **zero** whether it is placed in a **uniform** or **non-uniform electric field**.



# Characteristics of Gauss's Law

(vii) (b) If a closed body encloses a **charge q**. then total **flux**.




$\phi_E = (q/\varepsilon_0)$
(A)




$\phi_E = (q/\varepsilon_0)$
(B)




$\phi_E = (q/\varepsilon_0)$
(C)

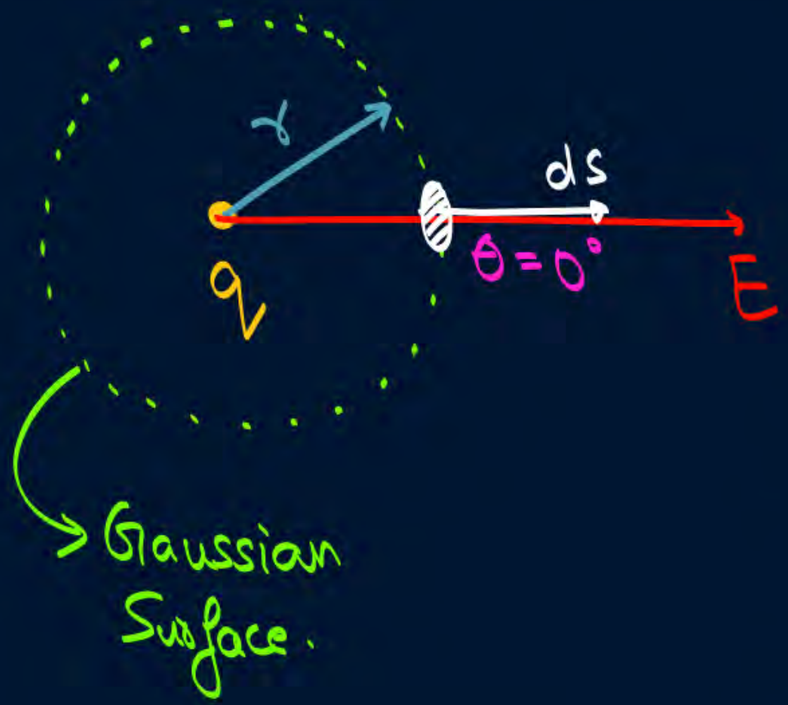



$\phi_E = (q/\varepsilon_0)$
(D)

linked with the body will be $\oint_S \vec{E}\cdot\vec{dS} = \frac{q}{\in_0}$. From this expression it is clear that the **flux** linked with a **closed body** is independent of the **shape** and **size** of the body and **position** of **charge inside** it. [See figures A, B, C and D].

# Electric Field due to a Point charge

$E = \frac{kq}{r^2}$




By Gauss's law, we can say that :-

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q_{enclosed}}{\epsilon_0}$$

$$\Rightarrow E \oint \vec{ds} = \frac{q}{\epsilon_0}$$

[Electric field at every point at distance r from the charge remains same i.e. E.]

$$\Rightarrow E(4\pi r^2) = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{(4\pi\epsilon_0)} \frac{q}{r^2}$$

$$\Rightarrow E = \frac{kq}{r^2}$$

# QUESTION

According to **Gauss's Law**, the total **electric flux** through a **closed** surface depends on:

A. Shape of the surface

B. Size of the surface

C. Total charge enclosed

D. Distribution of charges outside the surface.



# QUESTION

Using **Gauss's Law**, the **electric** field at a **distance r** from a point **charge q** is:

A $\frac{kq}{r}$

B $\frac{kq}{r^2}$

C $\frac{q}{4\pi r^2}$

D $\frac{q}{\varepsilon_0 r}$



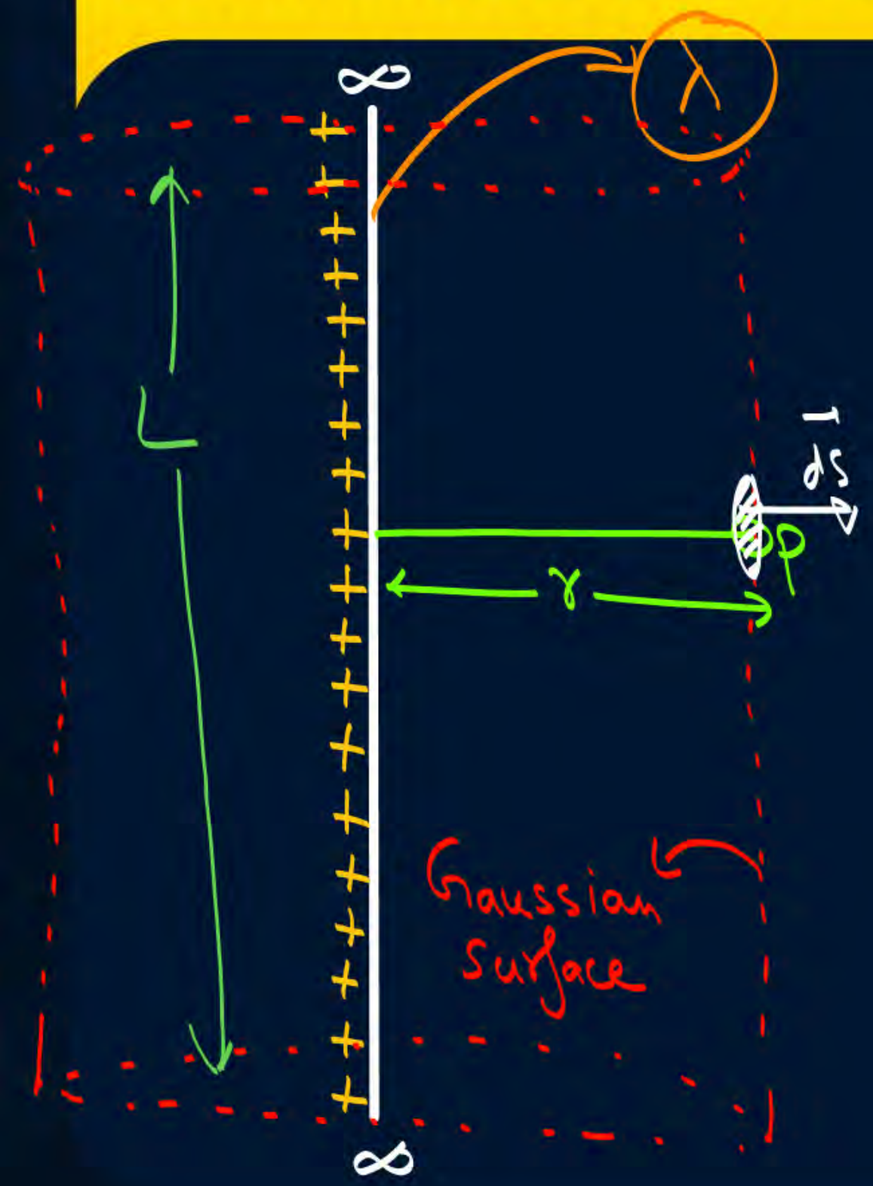
# Applications of Gauss's Law

- **Electric Field** due to an infinite linear charge distribution
- Electric Field due to a **shell of charge**.
- Electric Field due to a **Uniform sphere of charge.**
- Electric Field due to an infinite **thin non-conducting sheet.**



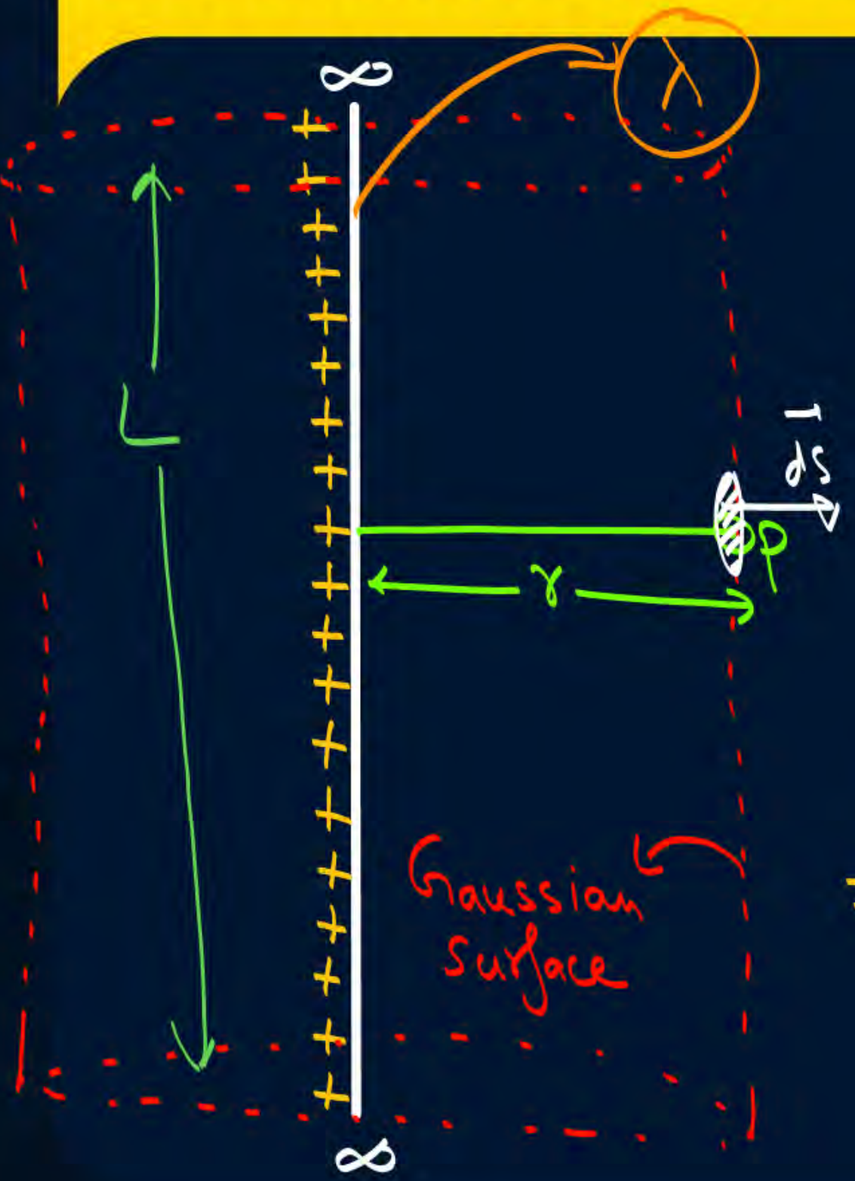



# Electric Field due to an infinite linear charge distribution




If we have an infinite linear charge distribution wire / rod, and we have to find the E.F. at point P at distance 'r' from the wire, then consider a Gaussian surface of length L, and radius r enclosing a part of the wire. Given that the linear charge density of the wire is λ.

# Electric Field due to an infinite linear charge distribution




↳ Linear charge density $= \lambda = \frac{Q}{L} \Rightarrow \{Q_{enc.} = \lambda L\}$

By Gauss's law: $\oint \vec{E} \cdot \vec{dS} = \frac{Q_{enc.}}{\epsilon_0}$

$\Rightarrow E \oint \vec{dS} = \frac{\lambda L}{\epsilon_0}$

$\Rightarrow E$ (Area of the cylindrical Gaussian Surface) $= \frac{\lambda L}{\epsilon_0}$

$\Rightarrow (E)(2\pi r L) = \frac{\lambda L}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r} \Rightarrow E = \frac{2k\lambda}{r}$

# DADA Method :

जो भी Shape charged body की होगी, उसके Gaussian Surface की भी Corresponding 3D shape होगी।

| ex: Body | GAUSSIAN SURFACE. |
| --- | --- |
| → Point charge | Sphere |
| → Wire/Rod | Cylinder |

# QUESTION

A point charge of **5 μC** is placed at the center of a spherical surface of **radius 0.1m**. The **electric flux** through the sphere is:
**($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$)**

**A** $5.65 \times 10^5\ Nm^2/C$

**B** $4.52 \times 10^5\ Nm^2/C$

**C** $2.82 \times 10^5\ Nm^2/C$

**D** $3.39 \times 10^5\ Nm^2/C$



# QUESTION

An **infinitely** long straight wire has a **uniform linear charge density** $\lambda$. Using **Gauss's** Law, the electric field at a perpendicular **distance r** from the **wire** is:

**A** $\frac{\lambda}{2\pi\varepsilon_0 r}$

**B** $\frac{\lambda}{4\pi\varepsilon_0 r}$

**C** $\frac{\lambda}{\pi\varepsilon_0 r}$

**D** $\frac{\lambda}{2\pi\varepsilon_0 r^2}$



# QUESTION

A thin, infinitely long line of charge has a **uniform linear** charge density of $3 \times 10^{-6} C/m$. Find the electric field at a perpendicular distance of **0.2 m** from the wire.

$(\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2)$

A $2.7 \times 10^5$ N/C

B $3.0 \times 10^5$ N/C

C $4.5 \times 10^5$ N/C

D $6.8 \times 10^5$ N/C

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$$= \frac{3 \times 10^{-6}\ C/m}{2 \times (3.14) \times 8.85 \times 10^{-12} \times (0.2)}$$

$$= 2.69 \times 10^5\ N/C$$



# THANK YOU

